Inference often combines conditioning and summarizing

Observed		**Condition**		**Summarize**
y	$\rightarrow$	$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z\mid y}, \Sigma_{z\mid y}\right)$	$\rightarrow$	$\hat{z} = E[z \mid y] = \mu_{z\mid y}$

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y)\, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z\mid y}$

A point estimate needs a loss function

Choose a reconstruction after observing y

$$m = E[z|y], \quad V = \mathrm{Cov}[z|y]$$

Choose any estimate $a \in \mathbb{R}^d$ to reconstruct the unknown vector $z \in \mathbb{R}^d$

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

Center the error at the conditional mean

$$r = z - m, \quad z - a = r + (m - a)$$

$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

- A reconstruction is judged by conditional expected squared error
- Given y, the chosen a is fixed; uncertainty remains in z

The conditional mean minimizes squared-error risk

Average the centered expansion Previous slide: $r = z - m$, $V = \mathrm{Cov}[z|y]$

$$E[r|y] = 0$$
$$E[2r^T(m-a)|y] = 2E[r|y]^T(m-a) = 0$$
$$E[\|r\|^2|y] = \sum_{j=1}^{d} V_{jj} = \mathrm{tr}(V)$$

Conditional squared-error risk

$$R(a|y) = \underbrace{\mathrm{tr}(V)}_{\text{Remaining uncertainty}} + \underbrace{\|m-a\|^2}_{\text{Excess risk from the chosen estimate}}$$

Optimal reconstruction

$$a^*(y) = m = E[z|y]$$
$$\min_a R(a|y) = \mathrm{tr}(V)$$

For the conditional Gaussian

$$m = \mu_{z|y}, \quad V = \Sigma_{z|y}$$

- This decomposition needs finite conditional second moments, not Gaussianity.
- The conditional mean minimizes model expected squared error; covariance describes the uncertainty that remains.

Murphy, Book 1 §5.1.5.1; vector decomposition: course elaboration

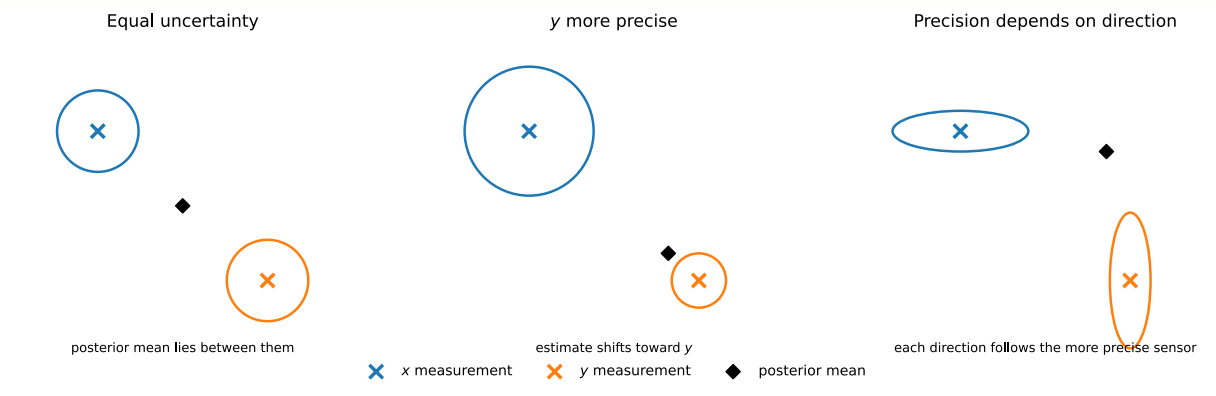
Sensor fusion combines measurements according to their uncertainty

Two noisy measurements of one state

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)\; x \mid z \sim \mathcal{N}(z, \Sigma_x), \quad y \mid z \sim \mathcal{N}(z, \Sigma_y)$$

Same two operations

$$p(z \mid x, y) \quad \rightarrow \quad E[z \mid x, y]$$



Course schematic after Murphy, Book 2 §2.3.2.3, Fig. 2.10; `sensor_fusion_figure.ipynb`

- Equal uncertainty puts the posterior mean between the measurements
- A more precise sensor pulls the posterior mean closer to its measurement
- With anisotropic uncertainty, the weighting can differ by direction

A point estimate needs a loss function

Carried forward Inference often combines conditioning and summarizing

$$\hat{z} = E[z|y] = \mu_{z|y}$$

Observed y; unknown $z \in \mathbb{R}^d$; $a \in \mathbb{R}^d$ fixed at this y.

Squared-loss criterion

$$\text{Assume } E[\|z\|^2|y] < \infty$$

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

Center at m

$$m = E[z|y], \quad V = \text{Cov}[z|y], \quad r = z - m$$

Earlier Gaussian: $m = \mu_{z|y}$, $V = \Sigma_{z|y}$.

$$z - a = r + (m - a)$$

$$\|z - a\|^2 = \|r\|^2 + 2r^T \cdot (m - a) + \|m - a\|^2$$

Murphy, Book 1 §5.1.5.1; vector decomposition: course elaboration

- We choose a to reconstruct z by minimizing posterior expected squared error $R(a|y)$.
- Centering writes the error as random residual r plus the fixed offset $m - a$.

The conditional mean uniquely minimizes squared-error risk

Carried forward Previous slide: m, V, and the pointwise squared-error expansion

$$m = E[z|y], \quad V = \mathrm{Cov}[z|y], \quad r = z - m$$
$$\|z - a\|^2 = \|r\|^2 + 2r^T \cdot (m - a) + \|m - a\|^2$$

Take conditional expectation; a, m are fixed at this y.

$$R(a|y) = E[\|r\|^2|y] + 2E[r^T \cdot (m - a)|y] + \|m - a\|^2$$

Centering cancels the cross-term

$$E[r|y] = E[z|y] - m = 0$$
$$E[r^T \cdot (m - a)|y]$$
$$= E[r|y]^T \cdot (m - a) = 0$$

Residual energy; $V = E[rr^T|y]$

$$E[\|r\|^2|y] = \sum_{j=1}^{d} E[r_j^2|y]$$
$$= \sum_{j=1}^{d} V_{jj} = \mathrm{tr}\, V$$

$$R(a|y) = \mathrm{tr}\, V + \|m - a\|^2$$

Substitute the two moments.

$$\|m - a\|^2 \geq 0 \quad \text{with equality iff} \quad a = m$$
$$a^*(y) = m = E[z|y], \quad \min_a R(a|y) = \mathrm{tr}\, V$$

Earlier Gaussian: $a^*(y) = m = \hat{z} = \mu_{z|y}$

$$\min_a R(a|y) = \mathrm{tr}\, \Sigma_{z|y}$$

- Risk is residual uncertainty plus the squared distance of the chosen action from the mean.
- This is model expected loss; finite conditional second moments suffice without Gaussianity.

Choose a reconstruction by its conditional squared-error risk

Observed y; choose $a \in \mathbb{R}^d$ to reconstruct unknown $z \in \mathbb{R}^d$

Squared-error criterion

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

Center at the conditional mean

$$m = E[z|y], \quad r = z - m, \quad z - a = r + (m - a)$$

$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

- Choose a to minimize conditional expected squared error $R(a|y)$.
- Given y, offset $m - a$ is fixed; uncertainty remains in r.

Centering the error makes the posterior mean optimal

Average the centered expansion Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

Take $E[\cdot\,|y]$; $m-a$ is fixed

$$E[r|y] = E[z|y] - m = 0$$
$$E\left[r^T(m-a)|y\right] = E[r|y]^T(m-a) = 0$$
$$V = \mathrm{Cov}[z|y]$$
$$E\left[\|r\|^2|y\right] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \mathrm{tr}\, V$$

Conditional squared-error risk

$$R(a|y) = \mathrm{tr}\, V + \|m-a\|^2$$
$$\|m-a\|^2 \geq 0 \quad \text{equality iff} \quad a = m$$
$$a^*(y) = m = E[z|y]$$
$$\min_a R(a|y) = \mathrm{tr}\, V$$

Earlier Gaussian

$$a^*(y) = \mu_{z|y}$$
$$\min_a R(a|y) = \mathrm{tr}\, \Sigma_{z|y}$$

- Risk is residual uncertainty plus squared distance from the posterior mean.
- The posterior mean minimizes model expected squared error; covariance measures what remains.

Centering the error makes the posterior mean optimal

Average; $m - a$ fixed Previous slide: pointwise squared-error expansion

$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

$$E[r|y] = E[z|y] - m = 0 \qquad V = \mathrm{Cov}[z|y]$$

$$E[r^T(m - a)|y] = E[r|y]^T(m - a) = 0 \qquad E[\|r\|^2|y] = \sum_{j=1}^{d} E[r_j^2|y] = \mathrm{tr}\, V$$

Conditional squared-error risk

$$R(a|y) = \mathrm{tr}\, V + \|m - a\|^2$$

$$\|m - a\|^2 \geq 0 \quad \text{equality iff} \quad a = m$$

$$a^*(y) = m = E[z|y]$$

$$\min_a R(a|y) = \mathrm{tr}\, V$$

$$\text{Earlier Gaussian:} \quad a^*(y) = \mu_{z|y}$$

$$\min_a R(a|y) = \mathrm{tr}\, \Sigma_{z|y}$$

- Risk is residual uncertainty plus squared distance from the posterior mean.
- The posterior mean minimizes model expected squared error; covariance measures what remains.

Centering separates residual uncertainty from estimation error

Average the centered expansion Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

Take $E[\cdot\,|y]$; $m-a$ is fixed

$$E[r|y] = E[z|y] - m = 0 \qquad V = \mathrm{Cov}[z|y]$$

$$E\left[r^T(m-a)|y\right] = E[r|y]^T(m-a) = 0 \qquad E\left[\|r\|^2|y\right] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \mathrm{tr}\, V$$

Conditional squared-error risk

$$R(a|y) = \mathrm{tr}\, V + \|m-a\|^2$$

- Risk is residual uncertainty plus squared distance from the posterior mean.

The posterior mean minimizes squared-error risk

Conditional squared-error risk Previous slide: residual uncertainty plus squared distance from the mean

$$R(a|y) = \operatorname{tr} V + \|m - a\|^2$$

Choose a to minimize $R(a|y)$

$$\|m - a\|^2 \geq 0 \quad \text{equality iff} \quad a = m$$

$$a^*(y) = m = E[z|y]$$

$$\min_a R(a|y) = \operatorname{tr} V$$

For the conditional Gaussian: $a^*(y) = \mu_{z|y}$

$$\min_a R(a|y) = \operatorname{tr} \Sigma_{z|y}$$

- The conditional mean minimizes model expected squared error.
- The posterior covariance describes uncertainty that remains after the mean action is chosen.

Marginalization keeps the matching blocks

$$x \sim \mathcal{N}(\mu, \Sigma)$$

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$

$$= \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$

$$p(x_2) = \mathcal{N}(x_2 \mid \mu_2, \Sigma_{22})$$

Marginal proof

Select x_1: $P = \begin{bmatrix} I & 0 \end{bmatrix}$, $\quad x_1 = Px$

By the affine rule:

$$x_1 \sim \mathcal{N}(P\mu, P\Sigma P^T)$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.76–2.77; affine rule: Book 1 §3.1.1, Eq. 3.5

- Marginalizing x_2 integrates over all its possible values
- Cross-covariance affects conditioning, not the marginal of x_1

Conditioning shifts the mean by an observed deviation

$$p(x_1 \mid x_2) = \mathcal{N}\left(x_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

Mean proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2} = x_1 - \mu_1 - B(x_2 - \mu_2)$$

$$E[r] = 0,$$

$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22}$$

$$= 0$$

(r, x_2) is jointly Gaussian:

zero cross-covariance gives independence

$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; proof uses affine Gaussian closure

- Conditioning shifts the mean according to the observed deviation $x_2 - \mu_2$, cross-covariance, and observed variance

Conditioning removes explained covariance

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

Covariance proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2}$$

From the mean proof: r and x_2 are independent

Murphy, Book 2 §2.3.1.3, Eqs. 2.78–2.79; proof reuses the independent residual

- Conditioning cannot increase variance in any direction
- In a Gaussian, this covariance depends on which variables are observed, not their values

Conditioning shifts the mean by an observed deviation

$$\mu_{1|2} = \mu_1 + B(x_2 - \mu_2)$$
$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

Mean proof

$$r = x_1 - \mu_1 - B(x_2 - \mu_2)$$
$$E[r] = 0$$
$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22} = 0$$

Affine closure: (r, x_2) is jointly Gaussian

Zero cross-covariance gives

r independent of x_2

$$E[r \mid x_2] = E[r] = 0$$
$$E[x_1 \mid x_2] = \mu_1 + B(x_2 - \mu_2) = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; course residual proof uses affine Gaussian closure

- The observed deviation $x_2 - \mu_2$ shifts the mean through $B = \Sigma_{12}\Sigma_{22}^{-1}$.

Conditioning shifts the mean by an observed deviation

Residual proof of the conditional mean

$$B = \Sigma_{12}\Sigma_{22}^{-1}, \quad r = x_1 - \mu_1 - B(x_2 - \mu_2)$$

$$E[r] = 0$$

$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22} = 0$$

Affine closure: (r, x_2) is jointly Gaussian

Zero cross-covariance gives r independent of x_2

$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_1 + B(x_2 - \mu_2) = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; course residual proof uses affine Gaussian closure

- The observed deviation $x_2 - \mu_2$ shifts the mean through $B = \Sigma_{12}\Sigma_{22}^{-1}$.

Conditioning shifts the mean by an observed deviation

Centered residual

$$B = \Sigma_{12}\Sigma_{22}^{-1}, \quad r = x_1 - \mu_1 - B(x_2 - \mu_2)$$
$$E[r] = 0$$
$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22} = 0$$

Affine closure: (r, x_2) is jointly Gaussian

Zero cross-covariance gives r independent of x_2

$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_1 + B(x_2 - \mu_2) = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; course residual proof uses affine Gaussian closure

- The observed deviation $x_2 - \mu_2$ shifts the mean through $B = \Sigma_{12}\Sigma_{22}^{-1}$.